EXCEL DASHBOARD PROJECT

BCA Student Performance Dashboard

A Microsoft Excel dashboard analyzing academic performance across all six semesters — pass rates, toppers, division trends, and back-paper tracking.

Prepared for Academic Review • Semester-wise Performance Analysis

Presented by Asrarul Haque

Objective & Purpose

Project Objective

The objective of this dashboard is to analyze the academic performance of BCA students across all semesters. It helps teachers and management identify student performance, pass percentage, topper details, and semester-wise results.

Who benefits

- Teachers — track individual & class performance
- Academic management — monitor pass rates & trends
- Administration — plan support for struggling students

SCOPE

6

Semesters Covered

91

Students Analyzed

100%

Excel-based (no external tools)

02 BUSINESS PROBLEM

Why This Dashboard Was Needed

Managing student performance manually is difficult because data is spread across multiple records. The college needed one central dashboard to answer key academic questions instantly.

1 Monitor Pass Percentage

Track how many students clear all subjects each semester

2 Identify Toppers

Recognize highest-performing students instantly

3 Compare Divisions

See distribution across First / Second / Third division

4 Analyze Semester Results

Spot trends across all six semesters

5 Identify Back Papers

Flag students needing remedial support

6 Track Overall Performance

Get one consolidated academic view

Dataset & Data Cleaning

Dataset Columns

- Student Name
- Roll Number
- Division
- Semester
- Marks
- Percentage
- Result
- Back Paper
- Total Marks

Data Cleaning Steps

- ✓ Removed duplicate records
- ✓ Checked blank values
- ✓ Standardized division names
- ✓ Corrected result values
- ✓ Verified total marks
- ✓ Organized semester-wise data

Excel Features Used

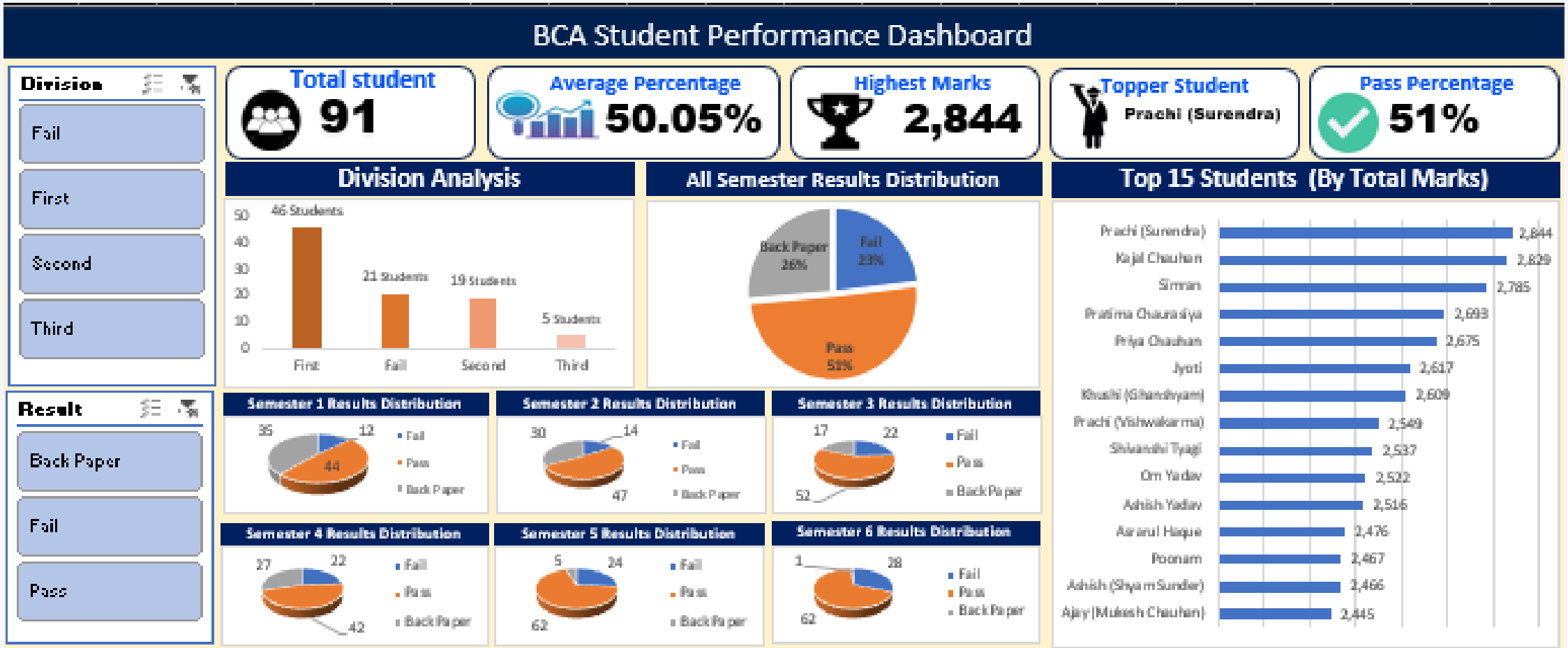
- Pivot Tables
- Pivot Charts
- Slicers
- Conditional Formatting
- Sorting
- Filtering
- Bar Charts
- Pie Charts
- Dashboard Formatting
- Cell Linking
- Basic Excel Formulas

KEY PERFORMANCE INDICATORS

Dashboard KPI Summary

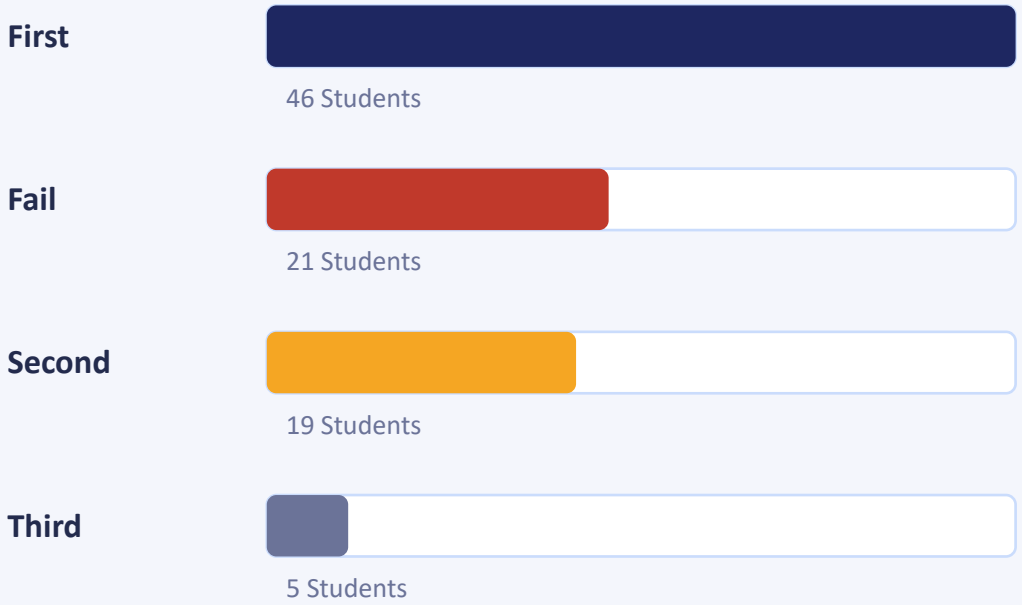
91	50.05%	2,844	Prachi (Surendra)	51%
Total Students	**Average Percentage**	**Highest Marks**	**Topper Student**	**Pass Percentage**
Total number of students in the dataset	Average academic performance of all students	Highest total marks scored by any student	Highest-performing student overall	Share of students who passed

Live Excel Dashboard Snapshot



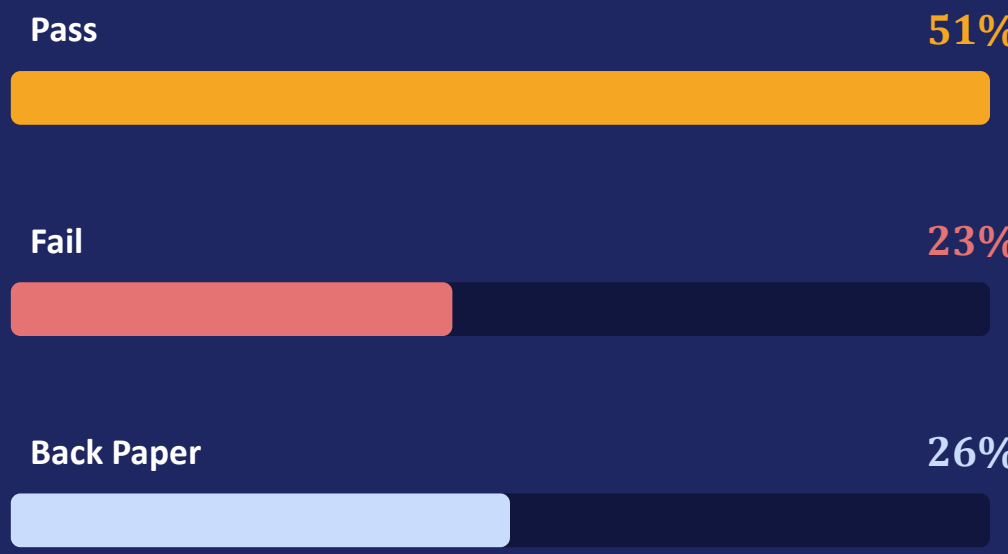
Division & Overall Result Distribution

Division Analysis



First Division has the highest share of students, reflecting strong overall academic performance.

All-Semester Result Distribution



Only 51% of students passed all subjects — nearly half carry at least one back paper or failed subject.

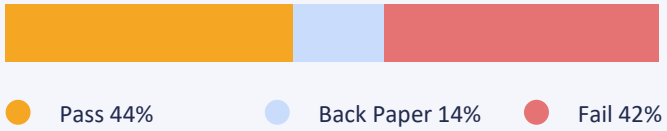
07 TOP PERFORMERS

Top 15 Students by Total Marks

1 Prachi (Surendra) 2,844 marks	6 Jyoti 2,617 marks	11 Ashish Yadav 2,516 marks
2 Kajal Chauhan 2,829 marks	7 Rivashi (Ghanshyam) 2,609 marks	12 Asrarul Haque 2,476 marks
3 Simran 2,785 marks	8 Prachi (Vishwakarma) 2,549 marks	13 Poonam 2,467 marks
4 Pratima Chaurasiya 2,693 marks	9 Shivanshi Tyagi 2,537 marks	14 Ashish (Shyam Sunder) 2,466 marks
5 Priya Chauhan 2,675 marks	10 Om Yadav 2,522 marks	15 Ajay (Mukesh Chauhan) 2,445 marks

Semester-wise Result Distribution

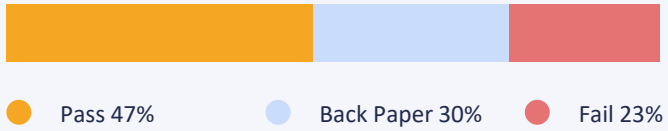
Semester 1



Pass 44% Back Paper 14% Fail 42%

Business Insight: highlights strongest and weakest results per semester

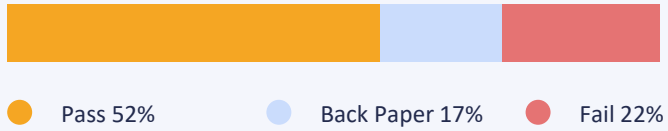
Semester 2



Pass 47% Back Paper 30% Fail 23%

Business Insight: highlights strongest and weakest results per semester

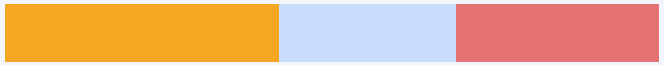
Semester 3



Pass 52% Back Paper 17% Fail 22%

Business Insight: highlights strongest and weakest results per semester

Semester 4



Pass 42% Back Paper 27% Fail 31%

Business Insight: highlights strongest and weakest results per semester

Semester 5



Pass 47% Back Paper 5% Fail 24%

Business Insight: highlights strongest and weakest results per semester

Semester 6



Pass 96% Back Paper 1% Fail 3%

Business Insight: highlights strongest and weakest results per semester

Key Insights

1. Total students analyzed: 91, across all six semesters.
2. Overall pass percentage stands at 51% — nearly half of students did not clear all subjects.
3. Prachi (Surendra) is the topper with 2,844 total marks.
4. First Division has the highest number of students, indicating a healthy performance base.
5. Semester-wise analysis pinpoints which semesters are most difficult for students.
6. The dashboard gives management a fast, visual way to support academic decision-making.

10 RECOMMENDATIONS

Business Recommendations

1. Provide extra support classes for students with back papers
2. Conduct remedial sessions for low-performing students
3. Reward top-performing students
4. Monitor semester-wise performance regularly
5. Track pass percentage every semester
6. Use the dashboard during academic review meetings

Challenges Faced & Future Improvements

Challenges Faced

- Cleaning inconsistent student data
- Creating dynamic Pivot Tables
- Linking slicers to multiple Pivot Tables
- Formatting the dashboard professionally
- Designing KPI cards

Future Improvements

- Attendance analysis
- Subject-wise performance
- Gender-wise analysis
- Department comparison
- Year-over-year trend analysis
- Power Query automation
- VBA refresh button

Thank You

This BCA Student Performance Dashboard uses Pivot Tables, Pivot Charts, Slicers, and Conditional Formatting to help academic management quickly identify high-performing students, analyze pass rates, and make data-driven decisions to improve student outcomes.

Excel Dashboard Project • BCA Student Performance Analysis

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